From Chatting with AI to Working with AI

Part 1 — From Clear Instructions to Verifiable Output

SPEAKER: **XINGYU YOU** DATE: **2026-09-18**

SECTION 0

From Chat to Reliable Work



1. How do we give AI better instructions?
2. How do we manage context before it becomes messy?
3. How do we turn a good AI interaction into a repeatable workflow — or an agent?

SECTION 1

From Vague Requests to Clear Instructions

MANAGER 15:34

Hi, could you please review the AE summary before tomorrow's deliverable review? Please check if there are any issues or inconsistencies. Thanks!

adsl.csv | adae.csv | t_ae_summary.rtf | analysis_spec.xlsx | program_log.txt

USER

Please review these files and tell me if there are **any issues**.

AI

OK, I read the AE table, subject-level data, AE data, specification, and program log. **The AE summary appears to have a Placebo issue**. The table shows Placebo N=6, but the safety population seems to be N=5 after excluding SUBJ012. **Some Placebo percentages and a few counts may need recalculation**, especially Any AE, Gastrointestinal disorders, and Nausea. Drug A looks consistent with the provided data. There may also be a PT ordering issue under Nervous system disorders, and the program log warning should be reviewed with the programmer because it may not fully match the table output...

Problem. The review scope is too broad.

"Any issues" does not define the required checks, including denominator, population filter, counts, percentages, sorting, and consistency.

Problem. The output format is unclear.

"Tell me" does not define a usable review structure with findings, evidence, corrected values, severity, and next steps.

Problem. The finding is not confirmed.

"Appears" does not confirm whether Placebo N=6 is wrong, what the correct value is, or which source supports it.

Problem. The impact is not quantified.

"Some" does not identify the affected rows or show their current and corrected values.

Problem. The action is not specific.

"Review with the programmer" does not specify what needs to be fixed, rerun, and reconciled.

SECTION 1

From Vague Requests to Clear Instructions

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USER

Context I am preparing for tomorrow's deliverable review of an AE summary table.
Task Review the AE summary and identify confirmed issues or inconsistencies.
Input Use `adsl.csv` and `adae.csv` as source data, `t_ae_summary.rtf` as the output, and `analysis_spec.xlsx` and `program_log.txt` for rules and warnings.
Rules
1. Verify denominators, AE counts, percentages, and SOC/PT ordering against the specification.
2. Use only file-based evidence; otherwise mark "Needs confirmation."
3. Count each subject once per SOC and PT, and separate confirmed findings from possible issues.
Output Start with a brief executive summary, then provide a review table with Finding, Severity, Evidence, Current/Corrected Value, Impacted Rows, and Next Step.

C — Context

T — Task

I — Input

R — Rules

O — Output

A good prompt externalizes the assumptions that normally stay in your head.
The goal is not to write more — it is to leave less for AI to guess.

USER

Identify subjects with **abnormal** lab results.

AI

...

SECTION 1

From Vague Requests to Clear Instructions

Define — Clarify the task

Generate — Create a first version

Review — Check the result

Refine — Fix specific gaps

Reuse — Save what works

Structure Before Prompting

Use when the task is still unclear. Ask AI to help identify missing information and turn rough ideas into a structured prompt.

USER

Please do not execute this task directly. First, check whether my prompt is missing background, inputs, constraints, output format, or validation criteria. Ask up to three most important questions. After I answer, produce an improved prompt.

Align Before Executing

Use when the prompt is mostly defined. Ask AI to confirm its understanding and execution plan before starting.

USER

Do not start the analysis yet. Please first tell me:
1. how you understand the goal of this task;
2. which files you plan to use;
3. how you will judge delays and risks;
4. whether anything is missing, conflicting, or unclear;
5. what steps you plan to follow.

Wait for my confirmation before you execute.

Correcting one misunderstanding now takes one sentence. Correcting it after execution may require redoing everything.

Common Problems

Not sure what is missing	Ask AI to diagnose the prompt first.
Too generic	Add criteria and priorities.
Wrong format	Provide an output schema.
Too long	Limit length and information hierarchy.
Wrong style	Provide an ideal example.
Off track	Remove irrelevant or outdated context.

SECTION 2

From Growing Context to Managed Context

USER

Make the summary shorter.

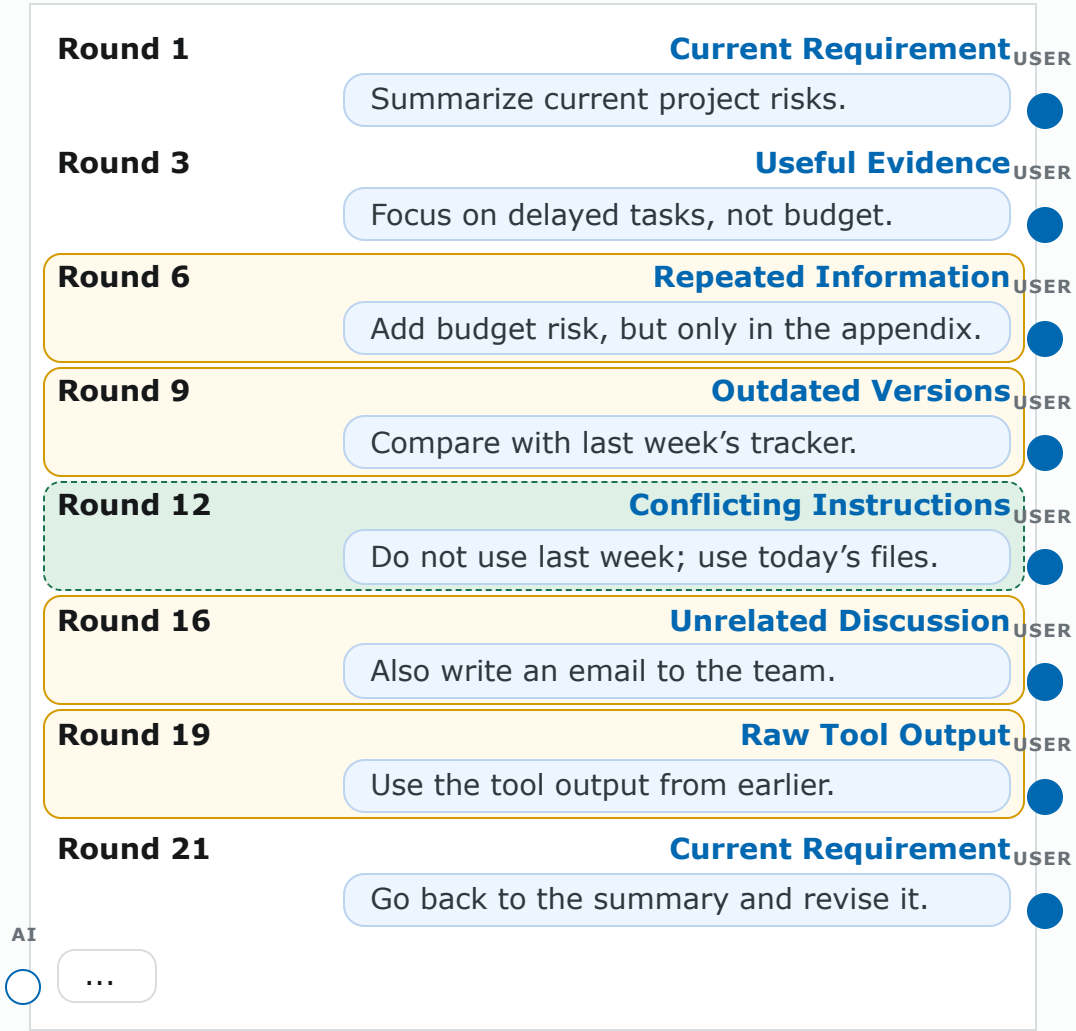
Context Container

System Instructions	Original Task	Conversation History	Uploaded Files
Previous Outputs	User Corrections	Tool Results	Current Requirements

A short message can still sit inside a much larger context. AI responds based on the context currently available to it, and different products may keep, retrieve, summarize, or truncate parts of the conversation history.

SECTION 2

From Growing Context to Managed Context



Common Problems

Uses old requirements	Mixes file versions	Ignores constraints
Budget analysis returns to the summary.	Last week's tracker is used by mistake.	The one-page limit disappears.

The problem is often not lack of information. It is that important information is not prominent enough.

Context Management

Remove Discard outdated requirements, rejected options, unrelated turns, and raw output after the useful result has been captured.

Old goals ***Old file versions*** ***Rejected options***
Unrelated content ***Conflicting outdated instructions***

Resolve When information conflicts, do not keep every version. Identify the current source of truth and mark older versions as superseded.

Compress Convert long history into a structured Working Context. **Reference:** Anthropic describes this as compaction and structured note-taking.

Useful history ***Earlier decisions*** ***Solved debugging***
Repeated discussion ***Long tool output***

Retrieve Keep the Working Context small and load files only when needed. **Reference:** Anthropic describes this as just-in-time context and progressive disclosure.

SECTION 2

From Growing Context to Managed Context

STOP method

S — Stop	T — Take Stock	O — Organize	P — Proceed Cleanly
Stop adding more requirements.	Summarize goal, progress, and issues.	Separate valid, outdated, and unresolved information.	Continue with a clean brief.

Reference: Anthropic emphasizes that when context becomes long, you should extract the actual state that needs to continue, including decisions already made, unresolved questions, current progress, and next steps, instead of simply summarizing the whole conversation.

Structured Handoff

Old Conversation

21 rounds
3 file versions
5 revised requirements
2 unrelated questions
Thousands of words

→

Goal	Finalize the one-page efficacy summary for the single-arm study.
Current State	The final dataset is `study_final_v3.csv`. The primary endpoint is response at Week 12, the analysis population is FAS, and the current analysis is complete.
Decisions	The summary should report N, responders, response rate, and 95% CI. It should use the prespecified response definition and remain limited to one page.
Open Issues	The confidence interval method has not been finalized. Whether to include sensitivity analysis is still undecided, and the final table format has not been approved.
Ignore	Ignore results from `study_v1.csv` and `study_v2.csv`, the old response definition, and earlier exploratory subgroup analyses.
Next Step	Revise the current efficacy summary. Do not make decisions on the open issues without confirmation.

In practice, keep this section concise. The wording is expanded here for slide readability.

SECTION 3

From AI Output to Verifiable Results

AI

Based on the project tracker, 24 tasks were reviewed. **Seventeen tasks were completed, four remained in progress, and three were overdue.** **System Testing had the largest schedule variance at six days.** **The action-item file contained two rows where the Owner field was missing.** **A comparison between the tracker and meeting notes also identified an inconsistent status for Task 17.** Additional analysis may be required to understand the causes of these issues...

The core of output design is not simply making text shorter. It is rearranging information hierarchy for the person who will use it.

	Programmer	Statistician	Leader
Audience	**Audience:** Statistical programmer responsible for QC resolution.	**Audience:** Statistician responsible for interpretation and consistency.	**Audience:** Project lead responsible for decisions and risk priority.
Purpose	**Purpose:** Resolve QC findings quickly and reproducibly.	**Purpose:** Confirm whether findings affect analysis conclusions.	**Purpose:** Understand current project risk and required decisions.
Priority	**Priority:** Location → discrepancy → impact → action.	**Priority:** Definition → population → estimate → uncertainty.	**Priority:** Risk → business impact → owner → next step.
Format	**Format:** Structured issue list with exact source references.	**Format:** Short review memo with assumptions and open questions.	**Format:** Snapshot plus decision-needed bullets.
Length	**Length:** ≤5 lines per issue.	**Length:** One concise paragraph per analysis point.	**Length:** Three key points, each under two sentences.
Exclusions	**Exclusions:** Do not infer root cause without table, dataset, variable, or log evidence.	**Exclusions:** Do not change statistical interpretation without confirmation.	**Exclusions:** Do not include technical calculation detail in the main summary.

USER

USER

USER

Output Repair Prompt

USER

Reorganize the verified analysis into a meeting-ready deliverable. Keep the numbers unchanged, lead with the most important findings, separate evidence from interpretation, move supporting detail to the appendix, and clearly mark anything uncertain or unsupported by the source data.

SECTION 3

From AI Output to Verifiable Results

AI

The Placebo percentage is incorrect **because the table program used the wrong population filter**.

Observed	Evidence	Supported Finding
Table: 12/40 (30.0%)	Analysis dataset: 12/38 (31.6%)	The denominator is inconsistent.

↓

⚠ **Unsupported Explanation**

The program used the wrong population filter.

Because we do know the denominator is inconsistent, but from the table and dataset alone, we do not know why. It may be the population filter, input dataset, macro parameter, derivation, or version issue.

FACT

Directly supported

The table uses N=40, while the specified analysis population in the analysis dataset contains N=38.

INFERENCE

Reasonable, but needs verification

The discrepancy may come from the population selection logic.

UNKNOWN

Cannot be determined from current evidence

The exact root cause has not been confirmed.

Statement	Type	Evidence	Confidence	Missing
Placebo denominator is inconsistent	Fact	Table / Dataset	High	—
Wrong population filter caused it	Unknown	No source	—	Program logic

Evidence Check Prompt

USER

Review each key claim and classify it as FACT, INFERENCE, or UNKNOWN based only on the available evidence. Treat a claim as FACT only when it is directly supported by the source data and cite that source; label it as INFERENCE when it is reasonable but not directly confirmed and state what still needs verification; use UNKNOWN when the available evidence is insufficient to determine the answer. Do not present inferences or unknowns as established conclusions.

Reliable AI work requires clear instructions, relevant context, and verifiable evidence.